The Internet Protocol (IP)

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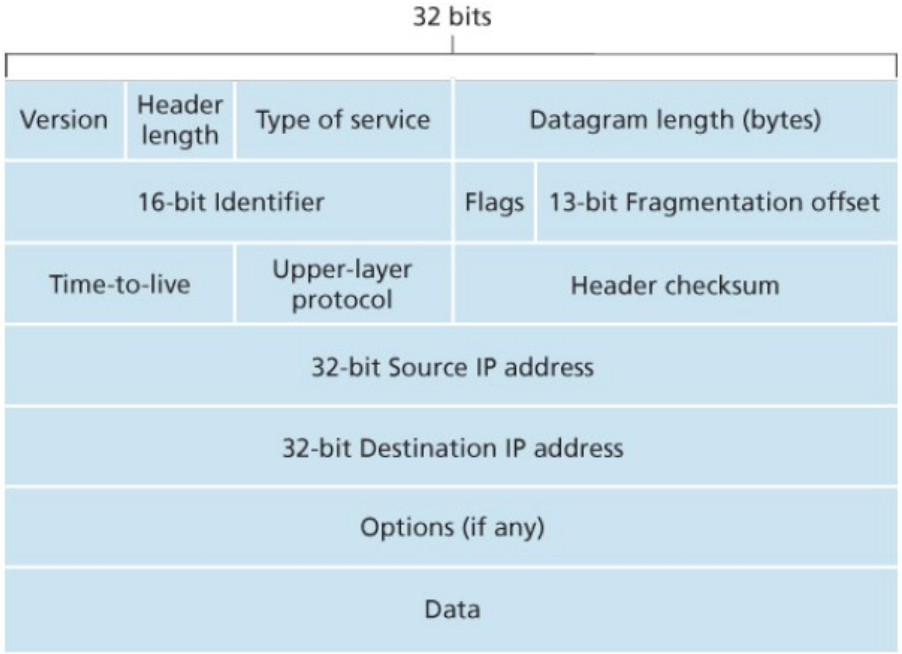
IP: The Internet Protocol

- The “IP” in “TCP/IP”, the Internet protocol suite
  - Internet Protocol version 4 (IPv4) – first major version, dominant on the Internet
  - Internet Protocol version 6 (IPv6) – in increasing deployment on public Internet since ~2006

- Task: deliver packets from source host to destination host based on *IP addresses* in the packet headers
- Defines:
  - Packet structures that encapsulate the data to be delivered
  - Addressing methods to label the datagram with source and destination information

IPv4 Datagram Format

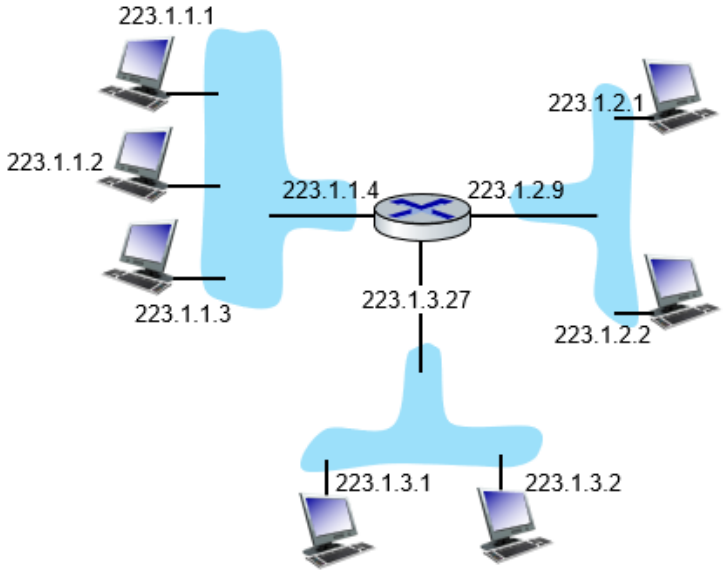



```
Internet Protocol Version 4, Src: 40.99.10.34, Dst: 192.168.0.147
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 40
    Identification: 0xe7b3 (59315)
  v 010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
        .1.. .... = Don't fragment: Set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 241
    Protocol: TCP (6)
    Header Checksum: 0xae5b [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 40.99.10.34
    Destination Address: 192.168.0.147
```

Fun with TCP/IP headers

IPv4 Addressing

- **IP address:** 32-bit identifier associated with each host or router *interface*
- **Interface:** connection between host/router and physical link
  - A host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)
  - A router typically has multiple interfaces



Dotted-decimal IP address notation:

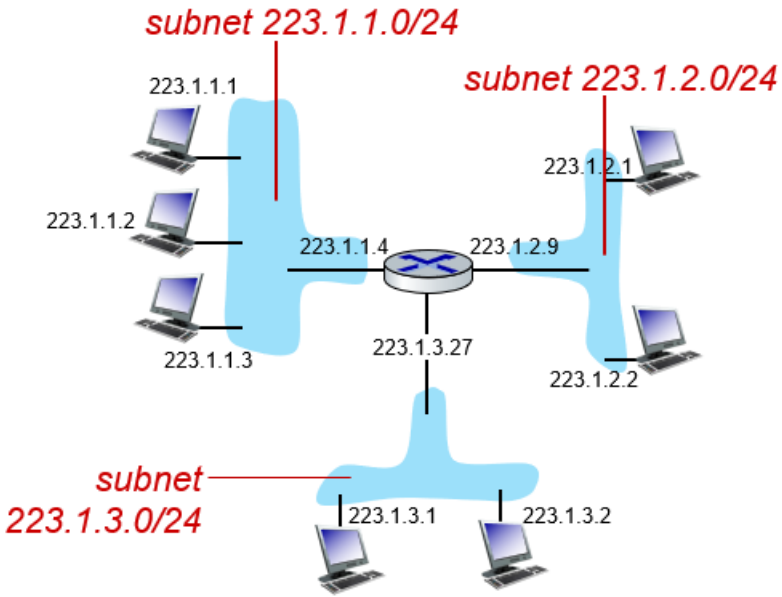
223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

Subnets

- **Subnet:** Device interfaces that can physically reach each other without passing through an intervening router
  - E.g., connected via LAN

- IP address structure:
  - *Subnet part:* devices in same subnet have common high order bits
  - *Host part:* remaining low order bits

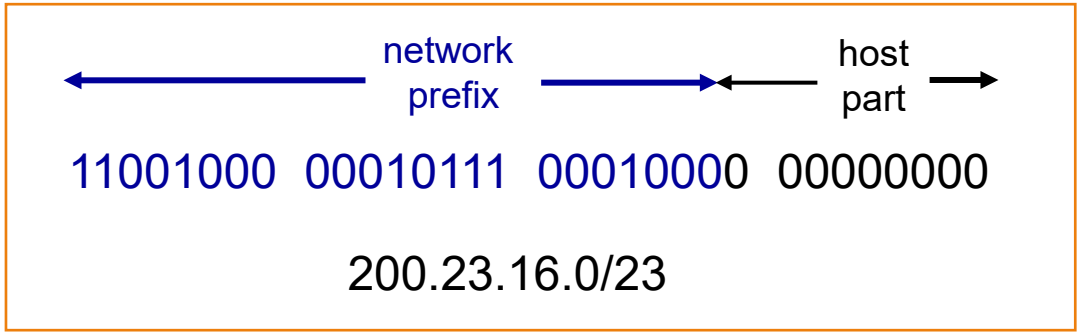


Subnet mask: /24

Means: subnet part is the 24 high-order bits

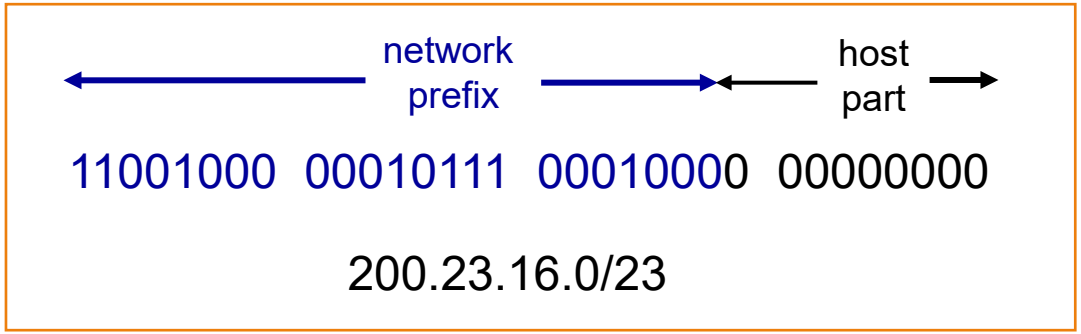
Classless Interdomain Routing (CIDR)

- CIDR: the Internet's address assignment strategy
  - Defined in RFC 4632
  - Generalizes the notion of subnet addressing
- Divide the 32-bit IP address into two parts: dotted-decimal form **a.b.c.d/x**
  - **x** indicates the number of bits in the first part of the address
  - The x most significant bits is called the *(network) prefix*



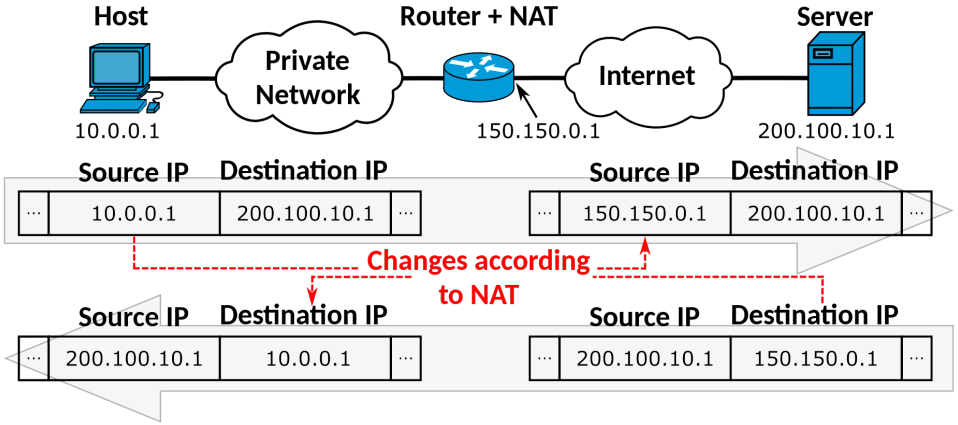
CIDR Usage

- An organization is typically assigned a block of contiguous addresses with a common prefix
  - Routers outside the organization only need to consider the x bits of network prefix
  - The remaining (32-x) bits distinguish devices within the organization



Network Address Translation (NAT)

- A way to map *multiple private addresses* inside a local network to *a public IP address*
  - Done by modifying network address information in the IP header of packets while they are in transit across a traffic routing device
- Has become popular for conserving global address space for IPv4



Special IPv4 addresses

`0.0.0.0`	Current host (self)
`255.255.255.255`	Broadcast on local network
Host part all ones	Direct broadcast to all hosts on the network E.g., on net `150.290/16`, address `150.290.255.255` broadcasts to all hosts on the network
Host part all zeros	Refers to the network itself
Network part all zeroes	Refers to the host on the local network E.g., on net `157.11.64.0/22`, address `0.0.1.6` refers to `157.11.65.6`
`0/8`	No addresses allowed here
`127/8`	Network prefix for loopback, e.g., for testing

`10/8`
`172.16/12`
`192.168/16`
	Reserved for private internets. Routers on the public Internet will discard packets addressed to any of these addresses.

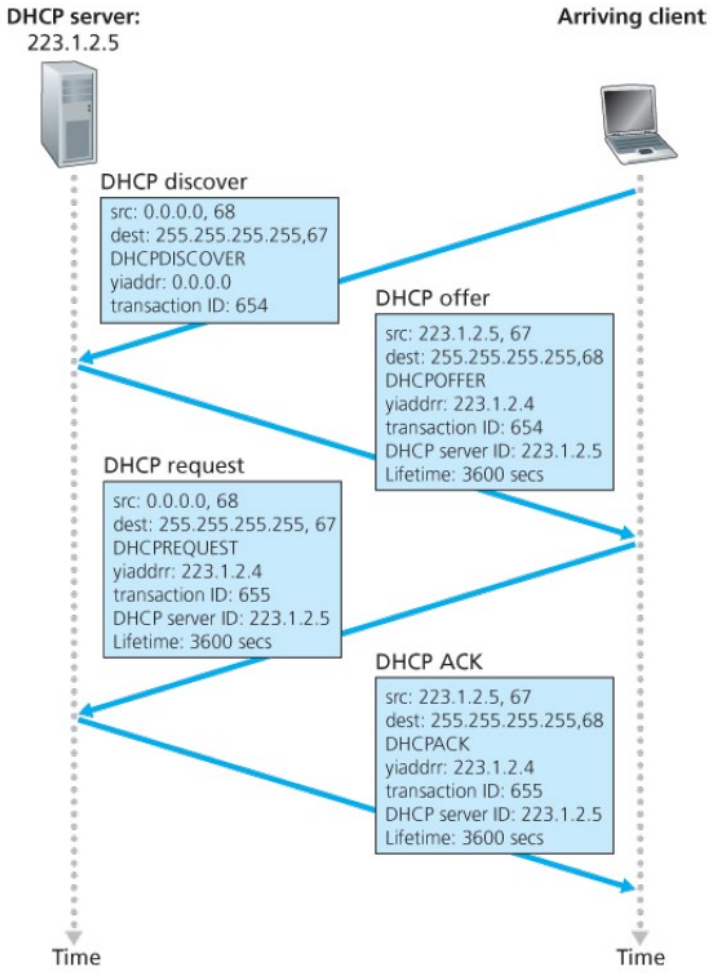
How to Get an IP Address?

Actually *two* questions:

- How does a network get IP address for itself (network part of address)?
  - ISP may provide to organizations from the block it has been allocated
  - Global authority: ICANN
- How does a host get IP address within its network (host part of address)?
  - Hard-coded by sysadmin in config file (e.g., /etc/rc.config in UNIX)
  - **DHCP (Dynamic Host Configuration Protocol):** dynamically get an address from a DHCP server (“plug-and-play”)

DHCP

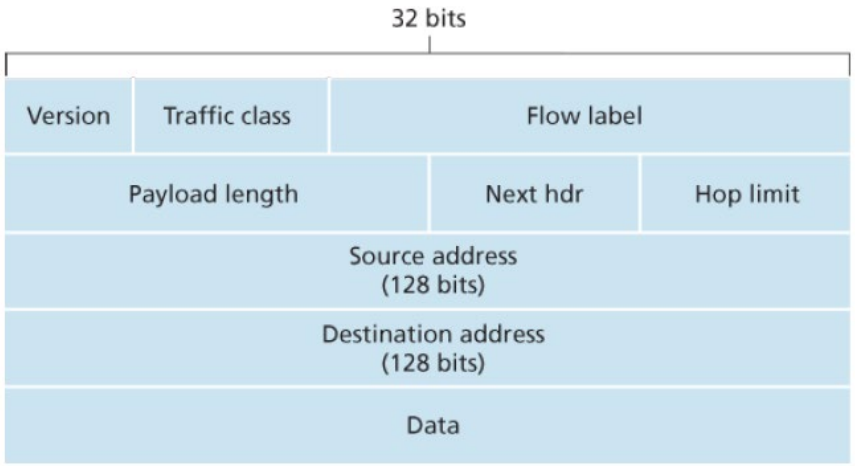
- A DHCP server maintains a pool of IP addresses to “lease”
- Host dynamically obtains IP address from network server when it “joins” network
- Allows reuse of addresses (only hold address while connected/on)
- Support for mobile users who join/leave network



IPv6 – Motivation

- Early 1990s – realization that 32-bit IPv4 address space is being used up at a breathtaking rate
- Response: new protocol: IPv6
  - Took opportunity to improve upon other IPv4 aspects
- Features
  - Larger address space, better management
  - Easier TCP/IP administration
  - Modern routing design
  - Better support for multicasting, security, mobility

IPv6 Datagram Format



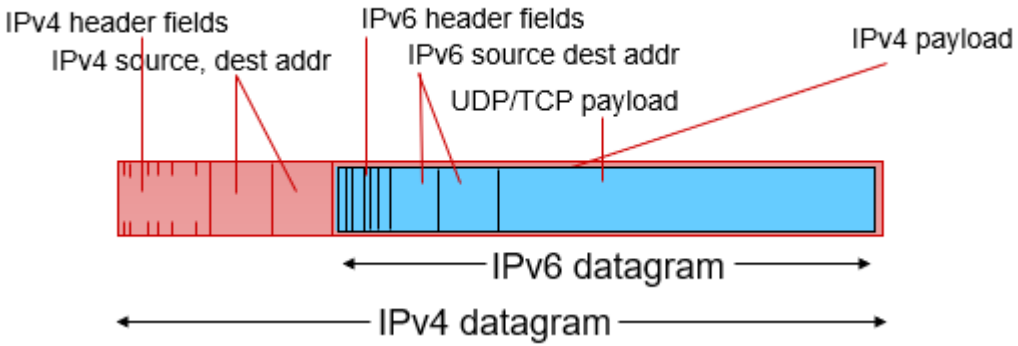
- Larger address space
  - Unicast, multicast, anycast
- Traffic class: for prioritizing
- Flow: not yet properly defined!
- No more fragmentation
- Next hdr ~ protocol in IPv4
- Hop limit ~ TTL in IPv4

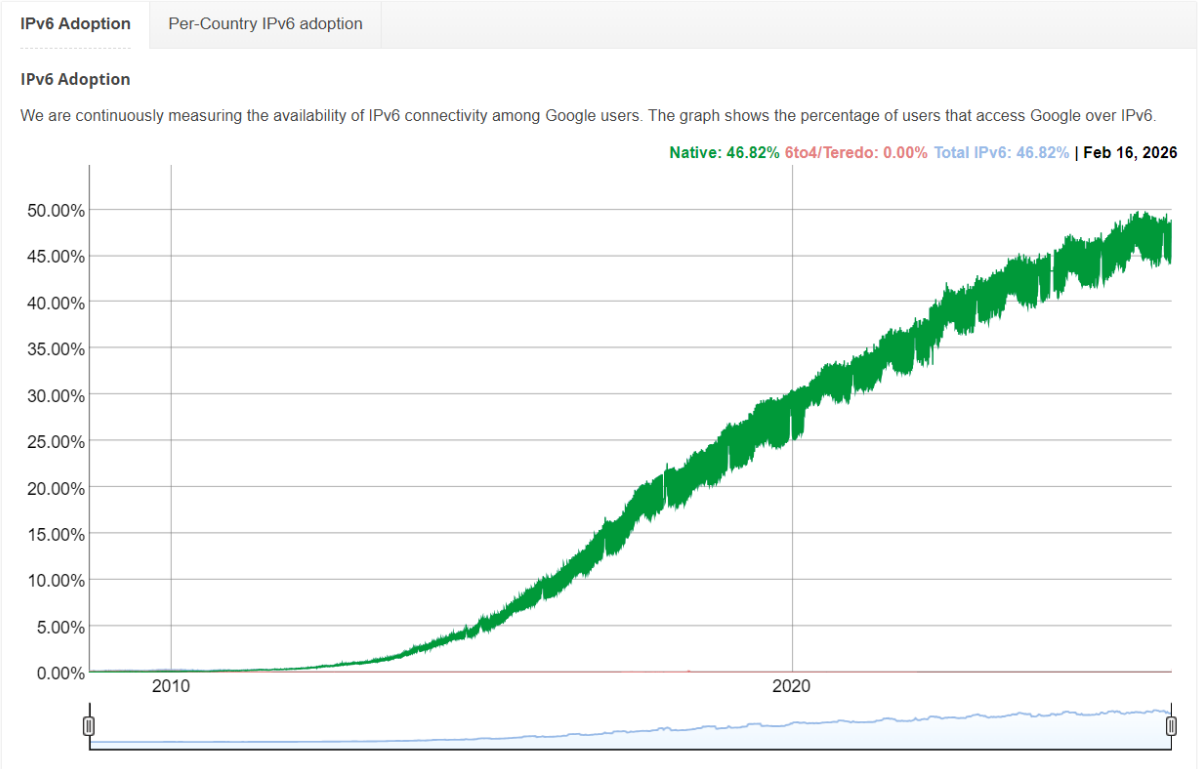
IPv6 Address Format

- 128 bits wide
- Convention: eight 16-bit hexadecimal, separated by colons
  - Leading zeros can be dropped
  - A single consecutive range of 0000 hextets can be replaced with ::
- Examples
  - 2001:0db8:85a3:0000:0000:8a2e:0370:7334
  - 2001:db8:85a3:0:0:8a2e:370:7334
  - 2001:db8:85a3::8a2e:370:7334
  - Loopback address: 0:0:0:0:0:0:0:1 (or ::1)

Transition from IPv4 to IPv6

- Not all routers can be upgraded simultaneously
- **Tunneling:** IPv6 datagram carried as payload in IPv4 datagram among IPv4 routers ("packet within a packet")



IPv6 Adoption

Statistics from Google: